01 / Acute renal colic

01 / Assessment and diagnosis

TRIAGE
Recognise red flags

IMAGING
Ultrasound / NCCT

LABORATORY
Urine + blood

Scope and how to read this sheet

Population	Adults with suspected stone-related acute flank pain. Pregnancy and children require adapted pathways; do not automatically apply adult medication advice.	Practical
Labels	Strong / Weak = EAU recommendation strength. EAU text = ungraded guideline narrative. Practical = editorial explanation or checklist. Section numbers refer to EAU Urolithiasis 2026.	Legend

1. Recognise the emergency first | 3.3.1; 3.4.2

Immediate imaging	Fever, a solitary kidney or an uncertain diagnosis warrants immediate imaging.	Strong
Emergency	Obstruction accompanied by UTI signs and/or anuria is a urological emergency. Do not delay analgesia or other urgent measures while arranging imaging.	EAU text
Initial assessment	Record vital signs, urine output and clinical stability at presentation. Ask about a solitary or transplanted kidney, previous stones/procedures, renal disease, pregnancy possibility, medication and allergies. A familiar pain pattern does not establish the diagnosis.	Practical
Differential	Assess competing causes according to presentation: pyelonephritis, vascular abdominal disease, appendicitis/diverticulitis, ectopic pregnancy or ovarian pathology, and testicular disease. This differential is a clinical reminder rather than an EAU diagnostic rule.	Practical

2. Image to answer the clinical question | 3.3.1

First imaging	Ultrasound is the initial imaging tool.	EAU text
Confirmation	After initial ultrasound, use non-contrast CT to confirm stone disease in acute flank pain.	Strong
Imaging record	Document side, stone size and position, obstruction and the contralateral kidney. Relate imaging to symptoms, laboratory results and baseline renal function. Specify which question remains unanswered if further imaging is requested.	Practical

3. Urine and blood | 3.3.2.c

Urine	Urine: dipstick for blood, white cells, nitrites and approximate pH; microscopy and/or culture.	Weak
Blood / coagulation	Blood: creatinine, uric acid, (ionised) calcium, sodium, potassium, blood-cell count and CRP. Obtain aPTT and INR when intervention is likely or planned.	Strong
Interpretation	Compare creatinine with previous results when available. Record sample timing and pending cultures. A urine result, imaging finding or pain response should not be interpreted in isolation.	Practical

01 / Acute renal colic

02 / Analgesia and urgent treatment

FIRST CHOICE
NSAID, assess risk

→

REASSESS
Response + safety

→

REFRACTORY PAIN
Drainage or URS

4. Analgesia: treat, reassess, escalate | 3.4.1

First choice	An NSAID is first choice, with cardiovascular risk and adverse effects considered.	**Strong**
Safety / alternatives	Use the lowest effective NSAID dose briefly; reduced GFR increases renal risk. Diclofenac requires cardiovascular/GI risk assessment. Adding an antispasmodic does not improve NSAID analgesia. Paracetamol is effective; opioids cause more vomiting, particularly pethidine.	**EAU text**
Second choice	Second choice in the acute setting: hydromorphone, pentazocine, tramadol or ketamine.	**Weak**
Prescribing / review	Choose formulation and route for the patient and clinical setting. Check allergies, interactions, renal function and the local product information before prescribing. Record the administered drug, dose, route, time, response and adverse effects; reassessment should determine the next step.	**Practical**
Dose context	For recurrent pain during expected spontaneous passage, EAU gives oral/rectal diclofenac sodium 100-150 mg/day for 3-10 days as an example. This is not an acute loading dose or a universal prescription.	**EAU text**
Local protocol	Other acute analgesic doses are not specified here: the cited EAU section does not supply a complete prescribing protocol. Use the local adult analgesia protocol and adjust to the individual patient.	**Practical**

5. Pain persists despite analgesia | 3.4.1

Escalation	Offer renal decompression or ureteroscopic stone removal for analgesia-refractory colic.	**Strong**
Review	Escalation means reassessing diagnosis, obstruction, infection and renal function as well as pain. Document why further medication, drainage or stone removal is being chosen. A pain-management failure should trigger a clinical decision, not an open-ended sequence of repeat doses.	**Practical**

6. Suspected infected obstruction | 3.4.2

Drainage	Sepsis with an obstructing stone requires urgent drainage by ureteral stent or percutaneous nephrostomy.	**Strong**
Antibiotics	Start antibiotics immediately; use intensive care if necessary. Reassess the antibiotic regimen against susceptibility results.	**Strong**
Cultures	Obtain blood and urine cultures alongside emergency management.	**EAU text**
After drainage	Collect urine again after decompression. Delay definitive stone treatment until sepsis has resolved.	**Strong**
Handover	Make the distinction explicit in handover: emergency drainage provides source control; definitive stone clearance is a separate treatment decision. Select empirical antibiotics through the local urosepsis protocol, considering prior cultures, allergies, renal function and resistance patterns.	**Practical**

01 / Acute renal colic

03 / Conservative care, review and handover

SELECT
Observation if suitable

MONITOR
Complications + review

CONFIRM
Passage + renal recovery

7. Observation and medical expulsive therapy | 3.4.3; 3.4.9

Stop / intervene	Stop conservative/MET treatment if infection, refractory pain or renal deterioration develops. Removal indications include unlikely passage, persistent pain, obstruction or renal insufficiency.	**EAU text**
Observation	Observation with periodic evaluation is an option for newly diagnosed small ureteral stones when active removal is not indicated.	**Strong**
MET	Offer an alpha-blocker as an option for distal ureteral stones 5-10 mm (3.4.3.a); this is off-label. Section 3.4.9.d.1 uses the broader wording distal stones >5 mm.	**Strong**
Shared plan	Explain what observation involves and document the agreed plan. MET is an option to discuss, not an automatic prescription for every stone or a substitute for treating an emergency. The cited recommendation does not specify a universal dose or treatment duration.	**Practical**

8. Before discharge or transfer | practical checklist

Clinical record	Document the working diagnosis, side, stone size/location, obstruction, renal function, infection assessment, symptom response and reason for the chosen setting of care.	**Practical**
Outpatient care	If outpatient care is selected, ensure a feasible medication plan, ability to take oral treatment, access to urgent help and an identified clinician/service responsible for review. These are practical discharge considerations, not a formal EAU discharge score.	**Practical**
Return instructions	Provide return instructions for fever/rigors, worsening or uncontrolled pain, persistent vomiting, reduced urine output or clinical deterioration. Give the patient a contact route and specify who will review pending results.	**Practical**
Passage / recovery	Filter urine for stone retrieval; confirm stone passage and recovery of baseline renal function.	**EAU text**
Stone analysis	Analyse a first stone using infrared spectroscopy or X-ray diffraction (3.3.2.c).	**Strong**
Review plan	Write down the review plan, required imaging/laboratory checks and the responsible service. This sheet deliberately does not invent a fixed EAU review interval. Recurrence prevention and metabolic assessment belong in their dedicated topic sheets.	**Practical**

9. Special populations | 3.3.3

Imaging	Pregnancy imaging: ultrasound first, MRI second, low-dose CT last. Children: ultrasound first; KUB or low-dose NCCT if insufficient.	**Strong**
Adapted care	Use the relevant pregnancy or paediatric pathway for treatment, imaging justification and dosing. A brief imaging hierarchy does not replace those pathways.	**Practical**

EAU Guidelines on Urolithiasis 2026, online chapter Guidelines, checked 28 October 2026. 3.3.1: imaging; 3.3.2.b-c: stone analysis/laboratory testing; 3.3.3: special-group imaging; 3.4.1: analgesia; 3.4.2: sepsis/anuria; 3.4.3.a: MET; 3.4.9.b-d.1: observation, escalation and MET.

Editorial draft v0.4. Independent educational summary, not an official EAU publication. Clinical review pending. Detailed procedure selection, sepsis treatment and metabolic prevention remain separate topics.